

Transfer Grilles: What Are They?



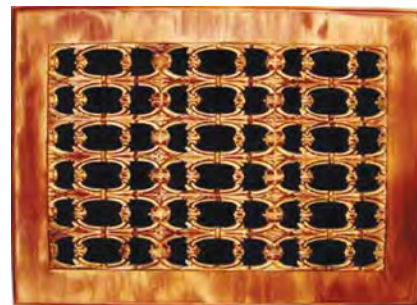
Doug Garrett is the president of Building Performance & Comfort, Incorporated, in Leander, Texas.

Q. I have a question for Doug Garrett, the author of “Beware the Closed Bedroom Door,” which appears on the *Home Energy* Web site, in the Do It Yourself section. Could you please explain transfer grilles and where I can purchase them?

**Jody Litten
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A. According to the Air Conditioning Contractors of America (ACCA) *Manual T: Air Distribution Basics*, “Transfer grilles can be used to combine isolated rooms or zones into a single larger return zone which can be served by a single return inlet.” In other words, they are used to connect rooms without return grilles to rooms (or zones) with return grilles so the air can easily flow back to the unit. Often this is a lot easier than running a return duct all the way back to the unit! To see why allowing this to happen is critical, read *Home Energy*’s past articles on the subject. (Search magazine archives at www.homeenergy.org.)

By the way, ACCA mentions that you could undercut the door by 1 inch for each 60 CFM. This would mean roughly a 6-inch undercut for a master suite with a 350 CFM flow! Um, I don’t think so; what else can you suggest?



Transfer grilles are used to connect rooms without return grilles to rooms (or zones) with return grilles so the air can easily flow back to the unit. (The grille above was used in Ancient Rome.)

Manual T recommends that transfer grilles be sized for a flow of 200 ft per minute (fpm) of face velocity. So if you have a master bedroom receiving 350 CFM of supply flow, you would install 1.75 ft² of gross grille area ($350/200 = 1.75$) or a 12 inch x 22 inch grille. The problem is that you can see and hear through a big grille like that. Not good for Mommy and Daddy’s bedroom, right?

Another method is to install a jump duct, which uses a short run of flex duct to connect one room to another. It usually goes from the ceiling of the isolated room to the ceiling of the room or zone with the return grille. Due to the very low static pressure across the duct, it usually takes an 8- or 10-inch duct to get enough flow, and the installation needs two boots and grilles, too. This is better for sight and sound attenuation, and I’ve done a lot of these.

Another method is to install Return Air Path (RAP) from Tamarack Technologies (<http://www.tamtech.com/rap.htm>). It comes in two parts, and it fits together to make a through-the-wall unit that looks like a high sidewall grille when installed. This innovative system can be easily used in new or retrofit applications and can provide for significant return flow. It has special stuff to provide good sound and sight attenuation, too. An easy and a cool solution! (Pun intended.) I hope this helps you out.

